OPERATIONS RUNBOOK

Standalone **WiFi**

Status panel · PKI, deploy, provisioning & lifecycle

Four staged, reversible cutover stages, then the three procedures that keep the certificates honest for as long as the panel hangs on the wall.

This runbook describes the proposed CONTRACT-SW v1.0 phase P5: issue the mTLS identities on chlorine, deploy the `solariPanel` listener on xenon, provision the panel over USB, and retire lithium from the chain. Every stage names its verification and its rollback before the next stage begins. Nothing in this document is standing consent — each stage is run deliberately, by the operator.

Status checked 21 October 2026. This is a proposed cutover procedure, not evidence that phase P5 was executed. Verify the currently installed panel daemon, listener, certificates, and lithium role before following it; live panel state was not checked for this edition.

Secrets discipline (red line). Key material lives on chlorine, in the panel's flash, and transiently on the provisioning bench under `0600` paths. It never enters git, never a handoff packet, never a log line. This document references every secret by path only.

VERSION	1.0.0	CONTRACT	CONTRACT-SW v1.0	CA	chlorine 10.7.0.10
DATE	2026-08-17	BRANCH	feat/standalone-wifi	LISTENER	panelgw:7443

SECTION 00

Contents & preconditions

Stages 1–4 run once, in order, each gated on its verification. Sections 05–07 are the recurring lifecycle procedures.

§	SECTION	WHAT IT DOES
01	Stage 1 · Issue identities	Server + client certificates from the chlorine step-ca
02	Stage 2 · Deploy the listener	solariPanel on xenon in listen+serial mode; DNS name
03	Stage 3 · Provision the panel	panelProv over USB; live-connection proof; secret cleanup
04	Stage 4 · Cutover	Panel on the wall on WiFi; lithium retired from the chain
05	Revocation	step-ca revoke + serial denylist + daemon restart
06	Renewal	Annual bench visit; daemon cert replacement; rollback window
07	Inventory & expiry watch	Cert inventory table; monthly expiry probe on xenon

Preconditions

- **Branch merged and flashed.** The standalone firmware and the listener-capable daemon come from `feat/standalone-wifi`; the flash follows the existing two-build byte-identical gate. This runbook assumes both are done.
- **chlorine step-ca reachable** at 10.7.0.10 and the operator can authenticate to it. Provisioner name: `[OPERATOR: provisioner]`.
- **DNS control** per the service-URL convention (SoR + BIND generator) to create `panelgw.akoria.net`.
- **A bench host** with a USB cable to the panel and the `status-panel/tools/panelProv` binary built (`make -C status-panel/tools`).
- **The battery pack record** in DESIGN-BRIEF-P4-POWER.md §1 is completed *before* the pack is ever connected (polarity confirmed against the silkscreen; the board has no charger and no power-path mixing).

Identity plan, fixed by contract. The daemon presents a certificate with SAN DNS `panelgw.akoria.net` and the `serverAuth` EKU; the panel presents SAN DNS `panel-01.panel.akoria.net` with the `clientAuth` EKU. Both chains anchor on the chlorine root. CN is display-only on both sides. The step-ca template must pin the EKU and SAN shape so a duplicate identity cannot be issued casually — that template pinning is an operational invariant of this runbook.

Port, not vhost. `panelgw.akoria.net` is a raw-TLS service on dedicated port **7443** — it is *not* an HTTPS vhost, so the usual :443 SNI convention does not apply. The name exists so certificates and firmware never carry a hard-coded address.

SECTION 01

Stage 1 · Issue the identities

Two certificates from chlorine: the listener's server identity and the panel's client identity. Run on a host with step authenticated against chlorine.

1.1 · SERVER CERTIFICATE (DAEMON ON XENON)

```
# on xenon, working dir /etc/solari-panel (root, umask 077)
step ca certificate "panelgw.akoria.net" panelgw.crt panelgw.key \
    --san panelgw.akoria.net
chmod 0600 panelgw.key && chmod 0644 panelgw.crt
step ca root ca.pem          # chlorine root, pinned by both sides
```

The certificate file must carry the full chain (leaf then intermediates) — the listener calls `SSL_CTX_use_certificate_chain_file`, and the firmware anchors on the root only. If your provisioner emits a bare leaf, append the intermediate(s) to `panelgw.crt`.

1.2 · CLIENT CERTIFICATE (THE PANEL)

```
# on the provisioning bench, working dir ~/panel-prov (umask 077)
step ca certificate "panel-01" panel-01.crt panel-01.key \
    --san panel-01.panel.akoria.net
```

1.3 · CONVERT TO DER FOR PROVISIONING

The credential store holds DER, not PEM. The client chain is leaf-then-intermediates concatenated DER.

```
openssl x509  -in ca.pem       -outform der -out ca.der
openssl x509  -in panel-01.crt -outform der -out panel-01.der
# if panel-01.crt contains intermediates, convert each and:
#   cat panel-01-leaf.der intermediate.der > panel-01.der
openssl pkey  -in panel-01.key -outform der -out panel-01.key.der
chmod 0600 panel-01.key panel-01.key.der
```

1.4 · VERIFY BEFORE ANYTHING IS INSTALLED

```
step certificate inspect panelgw.crt   # SAN panelgw.akoria.net, EKU serverAuth
step certificate inspect panel-01.crt  # SAN panel-01.panel.akoria.net, EKU clientAuth
openssl x509 -in panel-01.crt -noout -serial   # record in §07 inventory
openssl x509 -in panelgw.crt  -noout -serial   # record in §07 inventory
```

Both EKUs and both SANs must read exactly as above. A certificate missing its SAN or EKU will be refused by the peer at Stage 3 — fix it here, where it costs nothing.

> **Rollback.** Certificates are inert until installed. If either is wrong, revoke it (§05, no denylist entry needed for a never-deployed cert) and reissue. Nothing else has changed yet.

SECTION 02

Stage 2 · Deploy the listener

The one daemon binary gains a second transport. Serial mode on lithium keeps running untouched until Stage 4 — the fleet never loses its panel during this stage.

2.1 · DNS

Add `panelgw.akoria.net` as a CNAME to xenon's host record in the SoR, regenerate, and confirm: `dig +short panelgw.akoria.net` resolves to xenon's address.

2.2 · BUILD AND INSTALL ON XENON

```
make -C status-panel/daemon clean && make -C status-panel/daemon
make -C status-panel/daemon test           # must be green before install
sudo install -m 0755 status-panel/daemon/solariPanel /usr/local/bin/
```

2.3 · CONFIGURE

```
# /etc/solari-panel/solari-panel.conf on xenon (0600 root)
apiBase=https://dashboard.akoria.net/api
user=panel
passFile=/etc/solari-panel/panel.pass
serialDev=
listen=7443
listenAddr=[OPERATOR: xenon LAN address]
tlsCert=/etc/solari-panel/panelgw.crt
tlsKey=/etc/solari-panel/panelgw.key
clientCa=/etc/solari-panel/ca.pem
allowedSan=panel-01.panel.akoria.net
denylist=/etc/solari-panel/denylist.txt
```

- **Set `serialDev=` to an explicit empty value on xenon** — omitting the key leaves the default serial glob in place and the daemon quietly retries a device that will never appear. The empty value disables serial; that transport stays lithium's (bench fallback).
- **`listenAddr` is never a wildcard** — the daemon refuses `0.0.0.0`/`::` outright. Left unset, it selects the default-route address and logs the choice; pin it explicitly for production.
- **Create the denylist now, empty**: `touch /etc/solari-panel/denylist.txt`. A missing file logs "empty denylist" once; an unreadable file is fatal at start, by design.
- **File modes are enforced**: key `0600` or stricter (`0400` is fine), certs/CA `0644` or stricter (no group/other write, no exec), root-or-daemon-owned, or the listener refuses to start.

2.4 · START AND VERIFY

```
sudo systemctl enable --now solari-panel      # unit per lithium's, adjusted paths
journalctl -u solari-panel -n 20              # expect: TLS listener ready on <addr>:7443
# negative check from any LAN host — no client cert must mean refusal:
openssl s_client -connect panelgw.akoria.net:7443 </dev/null   # handshake fails; journal logs
#   "TLS client refused peer=... identity=... reason=..."
```

Rollback. `systemctl disable --now solari-panel` on xenon. The lithium serial chain has been running untouched throughout — the wall panel never noticed this stage.

SECTION 03

Stage 3 · Provision the panel

Credentials travel over USB only — PROVISION is rejected on WiFi by firmware rule. The tool is lockstep: one TLV, one PROVACK, resumable after any interruption.

3.1 · BENCH CONF

```
# ~/panel-prov/panel-prov.conf (0600) — secrets referenced by path
ssid=[OPERATOR: SSID]
pskFile=/home/[user]/panel-prov/psk.txt      # 0600; 8–63 bytes
serverHost=panelgw.akoria.net
serverPort=7443
caCert=/home/[user]/panel-prov/ca.der
clientCert=/home/[user]/panel-prov/panel-01.der
clientKey=/home/[user]/panel-prov/panel-01.key.der
ntpHost=[OPERATOR: gateway NTP, optional]
```

3.2 · RUN

```
./panelProv --dev /dev/serial/by-id/usb-Raspberry_Pi_Pico_* \
            --conf panel-prov.conf
```

Every item prints its PROVACK. ok throughout, ending with the commit ack, is success. The commit validates the whole staged set on-device (DER parse, key-matches-leaf, CA is a CA, host syntax) and writes the flash record atomically — the previous record stays valid at every interruption point. Common non-ok statuses: `commit-invalid` = staged set incomplete or fails validation (staging kept; fix the item and re-run); `crypto-invalid` = key/cert mismatch; `rejected-on-wifi` = you are somehow not on USB; `offset-mismatch` = auto-resumed by the tool.

3.3 · PROVE THE LIVE CONNECTION

- Power-cycle the panel with **no USB data connection** (power only, or battery).
- Panel obtains time (SNTP), joins WiFi, connects to the listener; xenon journal shows the mTLS accept with SAN `panel-01.panel.akoria.net`.
- Live fleet state renders within one poll cadence; a control-page command round-trips (queue → WiFi → STATE `lastCmdId`) within the ≤12 s bound.
- Plug USB back into a daemon-running host: firmware switches to USB ≤15 s; unplug: WiFi resumes via backoff.

3.4 · CLEAN THE BENCH

```
# ONLY after §3.3 is proven — this is the rollback window closing
shred -u psk.txt panel-01.key panel-01.key.der
```

Certificates (public) may be kept for the §07 inventory. Until the live connection is proven, the previous credential set — if this was a re-provision — stays staged aside on the bench as the rollback path.

Rollback. `panelProv --wipe` returns the panel to unprovisioned (WiFi off, USB serial exactly as before); or re-run panelProv with the retained previous set. USB serial always works regardless.

SECTION 04

Stage 4 · Cutover & retire lithium

The panel goes back on the wall on WiFi power-only. Lithium leaves the chain but keeps the binary — a bench is only ever one USB cable away.

4.1 · HANG THE PANEL

- Mount with USB **power only** (or the wall supply). Within the backoff cap the panel is rendering live state via panelgw.
- Watch one full rotation and the control page's freshness chip; both must read current.

4.2 · RETIRE LITHIUM FROM THE CHAIN

```
# on lithium
sudo systemctl disable --now solari-panel
# binary and config stay installed — bench fallback, not deletion
```

Nothing else on lithium changes. Plugging the panel into lithium and restarting the unit restores the full serial chain at any time — that is the standing fallback for firmware work, reprovisioning, and any WiFi outage that outlasts patience.

4.3 · POST-CUTOVER ACCEPTANCE

CHECK	PASS LOOKS LIKE
AP bounce	Restart the serving AP; panel reconnects unattended within the 60 s backoff cap + one poll
Server restart	Restart solari-panel on xenon; same reconnect bound; no stale-command replay (lastCmdId evidence)
Power pull	On battery: brightness caps, glyph shows, alarm arms with a 0xC-range episode, button acks; restore clears it
USB arbitration	USB into a daemon host wins ≤15 s; unplug resumes WiFi; no duplicate command application
Staleness	LINK LOST appears only on genuine snapshot staleness, never on a clean transport switch

Ride-through numbers. Measured draw and runtime at capped brightness with WiFi associated: `[OPERATOR: from the D19 measurement session]`. The ratified minimum ride-through: `[OPERATOR]`. No number is asserted here ahead of measurement, by contract.

Rollback. Take the panel down, USB into lithium, `systemctl enable --now solari-panel` on lithium, and optionally disable the xenon listener. That is the exact pre-P5 world.

SECTION 05 · 06

Revocation & renewal

One device, one issuer: a serial denylist instead of CRL machinery. Freshness is this procedure, run promptly — not a protocol guarantee.

05 · REVOKE A PANEL CERTIFICATE

```
# 1 — read the serial (from the §07 inventory, or the cert itself)
openssl x509 -in panel-01.crt -noout -serial      # serial=ABCD1234...
# 2 — revoke at the CA (on a step-authenticated host)
step ca revoke [serial]
# 3 — deny it at the listener (on xenon) — hex, one per line, # comments ok
echo 'ABCD1234...' | sudo tee -a /etc/solari-panel/denylist.txt
sudo systemctl restart solari-panel
# 4 — verify: a connect attempt with the old cert is refused AFTER chain
#     verification; the journal logs "TLS client refused peer=... identity=...
#     reason=serial denylisted"
```

Step 3 is the one that actually locks the door — the listener loads the denylist at start only. The CA revocation (step 2) is bookkeeping that keeps chlorine's records true and protects any future verifier that does consult the CA. Do both, in this order, same sitting.

Compromise posture. If the panel's key is suspected compromised (board stolen off the wall), revoke + denylist first, then reissue and reprovision at the bench. The credential store is plain flash by accepted threat model — physical possession of the board means possession of its key.

06 · RENEW (ANNUAL BENCH VISIT)

- **Window.** Renew when the §07 probe first alerts (<30 days to expiry). The alert lead time *is* the renewal window.
- **Panel cert:** Stage 1 §1.2–1.4 (fresh key + cert, new serial), then Stage 3 in full — panelProv, live-connection proof, *then* shred. The old set stays staged aside until the new set proves live; rollback is re-running panelProv with it.
- **Daemon cert:** Stage 1 §1.1 on xenon, install over the old chain, `systemctl restart solari-panel`. The panel's trust is unaffected — same root.
- **Old serials** of cleanly-replaced certs need no denylist entry; denylist is for revocation, not rotation. Update the §07 inventory either way.
- **On-device renewal (ACME) is out of scope** by contract — do not improvise it.

Rollback (renewal). Panel side: reprovision with the retained previous set. Daemon side: reinstall the previous chain + restart. Both are two-minute operations while USB serial keeps the wall alive.

SECTION 07

Inventory & expiry watch

The runbook is the system of record for what was issued. The probe exists so the renewal window opens itself.

7.1 · CERTIFICATE INVENTORY

IDENTITY	SERIAL (HEX)	NOTAFTER	ISSUED
panelgw.akoria.net	[OPERATOR]	[OPERATOR]	[OPERATOR]
panel-01.panel.akoria.net	[OPERATOR]	[OPERATOR]	[OPERATOR]

Fill at Stage 1 §1.4; strike and re-enter on every renewal. Two rows is the whole fleet — if a third row ever appears, the allowedSan template policy conversation happens first.

7.2 · EXPIRY PROBE (XENON, MONTHLY)

```
# /usr/local/sbin/panel-cert-expiry.sh (root, 0755)
#!/bin/sh
warn_days=30; rc=0
for c in /etc/solari-panel/panelgw.crt /etc/solari-panel/panel-01.crt; do
  if ! openssl x509 -in "$c" -noout -checkend $((warn_days*86400)); then
    logger -p daemon.warning -t panel-cert "expires <${warn_days}d: $c"
    rc=1
  fi
done
exit $rc
```

```
# crontab (root on xenon) — first of the month, 08:15
15 8 1 * * /usr/local/sbin/panel-cert-expiry.sh
```

- Keep a copy of the panel's **public** cert at `/etc/solari-panel/panel-01.crt` (0644) purely so the probe can watch both expiries from one host. It is not used by the listener.
- A TLS probe against :7443 was rejected by design — it would conflate reachability with validity. This probe reads the files.
- Wire the `daemon.warning` line into the existing alert path: [OPERATOR: journal-watch rule].

7.3 · STANDING INVARIANTS

- **Template pinning:** the chlorine provisioner template pins EKU + SAN shape for both identities; issuance outside the template is a policy change, not a routine op.
- **One session:** the listener accepts one authenticated session; newest wins. A second panel is a contract change (multi-panel is out of scope).
- **Provisioning is USB-only, forever.** Any future "provision over the network" idea reopens the contract; the firmware rejects it today with `rejected-on-wifi`.